



May 2025

Duration: 3 Hours

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

9. The baud rate is 2400 during serial communication using timer1 in mode 2, the value to be loaded in TH1 is. XTAL=11.0592 MHz
- A) FA B) E3
C) FD D) F4
10. In 8051 microcontroller, Timer operating in mode 0, the timer register size is
- A) 8 bit B) 13 bit
C) 18 bit D) 64 bit
11. How many interrupts in 8051?
- A) 4 B) 6
C) 5 D) 7
12. What is the value of the TH0 and TL0 register to be loaded to generate a delay of 0.5m sec using timer in Mode 1 in 8051. Assume Fosc=11.0592MHz.
- A) TH0=FA and TL0=33H B) TH0=33H AND TL0=FEH
C) TH0=FEH AND TL0=33H D) TH0=65H AND TL0=07
13. If the total machine cycle required is 23000 then the counter value required for time1 in mode 1 is
- A) 28A6 B) A628h
C) 68A6 D) FFFF
14. The external hardware interrupt in 8051 are
- A) INT0 and INT1 B) TF0 AND TF1
C) POWER UP RESET D) SERIAL RECEIVE/ TRANSMIT
15. ARM processor has
- A) 31 general purpose register B) 16 general purpose registers
C) 8 general purpose register D) 24 general purpose register
16. What is result in the register r0 after the execution of RSB r0, r1, #0
If content in register before execution are
r0 = 0x00000000
r1 = 0x00000077
- A) r0= 0xfffff89 B) r0= 0x00000777
C) r0= 0x00000077 D) r0 = 0x000000ff
17. ARM instructions are
- A) 8 bit in size B) 64 bit in size
C) varying size D) 32 bit in size
18. Thumb instruction mode in ARM processor uses
- A) 16 bit code size B) 32 bit code size
C) 8 bit code size D) 24 bit code size
19. What is the result after the execution of following instruction in r0?
MLA r0, r1, r2, r3
If r0 = 0x00000001
r1 = 0x00000002
r2 = 0x00000003
r3 = 0x00000004
- A) r0 = 0x0000000a B) r0 = 0x00000010
C) r0 = 0x0000000c D) r0 = 0x00000024
20. Which one of the following is valid for ARM for loading an immediate data in to a registers
- A) Most of the immediate values are loaded using barrel shifter B) instructions itself frames 32 bit immediate vale
C) No immediate data is valid in ARM D) A separate 32 bit data is subsequently stored in the program memory along with 32 bit opcode

PART - B: DESCRIPTIVE ANSWER QUESTIONS

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) With a neat diagram, explain the architecture of the 8051 microcontroller. Describe its key components and highlight their roles in microcontroller operations.	8	L*2	1	1
	b) Write an 8051 assembly language program to reverse the order of 10 bytes stored in consecutive memory locations starting from 50H. The reversed sequence should be stored in consecutive memory locations starting from 60H.	8	L3	1	1
2.	a) With an example explain four addressing modes of 8051 microcontroller.	8	L2	1	1
	b) Write an 8051 assembly language program to check whether a number stored at memory location 40H is even or odd. If the number is even, store 00H at memory location 50H; if the number is odd, store 01H at memory location 50H.	8	L3	1	1
3.	a) Briefly describe the RAM memory organization of 8051 microcontroller.	08	L2	1	1
	b) Write an 8051 ALP to convert a packed BCD number stored at memory location 50H into two ASCII digits. The converted ASCII values should be stored in R5 and R6 respectively.	08	L3	1	1
		Unit – II			
4.	a) Draw the internal port structure of 8051 microcontroller. Assume that two switches S1 and S2 are connected to P3.2 and P3.3 of 8051 microcontroller. When S1 is closed microcontroller must perform up count once and when S2 is closed microcontroller must perform down count once. Eight LEDs are connected to port 0 to display the count. Draw the interfacing diagram. Write an 8051 assembly code to illustrate the counter.	08	L3	3	1
	b) Describe the function of each bit in the following 8051 SFRs: i) TCON ii) TMOD iii) SCON iv) IE	08	L2	2	1
5.	a) Design a circuit to interface 4 seven segment displays using only 12 I/O lines of 8051 microcontroller. Write a program to display "2025" on the seven segment. Consider P2 to control the Segment and P3 to be used as Data lines for the segments.	08	L3	3	1
	b) With a Neat Diagram, Explain the Data Read and Data Write Operations of Port P1 in the 8051 Microcontroller.	08	L2	2	1
6.	a) Write an 8051 code in C language to perform 8-bit binary UP COUNT and DOWN COUNT. Also display the count values on LED	08	L3	3	1
	b) Explain briefly the interrupts of 8051, indicate their vector addresses. Write the steps to be executed when interrupt occurs.	08	L2	2	1

Unit – III

7.	a)	Explain the structure of 5 stage pipeline with an example in ARM core.	8	L2	5	1
	b)	Explain the functionality following instruction with an example for each.				
		i. MLA				
		ii. BIC				
		iii. RSB				
		iv. MVN	8	L2	4	1
8.	a)	Explain the principle of crossbar switch barrel shifter.	8	L2	4	1
	b)	Explain the multiplier architecture of an ARM processor with the help of a diagram. Additionally, describe the design of a multiplier with an example.	8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
